

**DEVELOPMENT OF A JOB MARKET ANALYTICS SYSTEM FOR COMPUTER
SCIENCE GRADUATES**

A FINAL YEAR PROJECT

SUBMITTED TO

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BY

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UNDER THE GUIDANCE OF

MR. ONEME LUCKY OGUME

IN PARTIAL FULFILMENT OF THE REQUIREMENTS

FOR THE AWARD OF THE DEGREE OF

BACHELOR OF SCIENCE (B.Sc.)

IN

COMPUTER SCIENCE

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DECLARATION

I, Ugwumba-Nwoye Michael (AU202200749), hereby declare that this project titled “Design and Implementation of Job Analysis System for CS Graduates” was carried out by me under the supervision of Mr. Oneme Lucky Ogume. This work is submitted in partial fulfilment of the requirements for the award of the Bachelor of Science (B.Sc.) degree in Computer Science at Augustine University, Ilara-Epe.

This project is my original work and has not been submitted elsewhere for any academic award. All materials used from other sources have been properly acknowledged through references.

Student's name and signature

Date

CERTIFICATION

This is to certify that the project titled “Design and Implementation of a Job Analysis System”, carried out by Ugwumba-Nwoye Michael with matriculation number AU202200749, of the Department of Mathematical Sciences, Augustine University, Ilara-Epe, has been read and approved as meeting the requirements for the award of the Bachelor of Science (B.Sc.) degree in Computer Science.

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Date

Date

DEDICATION

This project is dedicated first and foremost to God Almighty, whose grace, strength, and guidance made the completion of this work possible.

I also dedicate this project to my beloved parents, Mr. and Mrs. Ugwumba-Nwoye, for their unwavering support, encouragement, prayers, and sacrifices throughout my academic journey. Their belief in me provided the motivation needed to overcome challenges and remain focused on achieving my goals.

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ABSTRACT

.CS Jobber is a web based Job Market Analysis System for Computer Science graduates to understand the job market. The system is built by using HTML5, CSS3, JavaScript, Node.js and SQLite. It analyses job postings from 10,345 companies across the world and uses primary data collected from 20 Nigerian employers and 11 CS graduates to uncover skill gaps, salary trends and the demand for roles across 8 interactive pages.

The system was tested and evaluated against four objectives, all of which were met. The highest paying role is AI Engineer with the average salary of \$139,945. All five tracked skills show up in around 50% of listings. Networking is the most sought after skill by employers in Nigeria. CS Jobber is a real life data-driven career decision making tool for the graduates of Augustine University Ilara-Epe.

CHAPTER ONE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The job market for Computer Science graduates has become increasingly competitive and dynamic in recent years. As technology continues to advance rapidly, employers are constantly updating their requirements and expectations from computing professionals. Every year, a large number of Computer Science graduates enter the workforce, yet many struggle to secure employment due to a significant mismatch between the skills acquired during their academic training and the skills that employers actually demand (Ibrahim et al., 2024). This gap between academic preparation and industry expectations has become a growing concern for graduates, universities, and employers alike.

Currently, most Computer Science graduates rely on manual and unstructured methods to understand the job market. Graduates are required to independently search through multiple job listing websites, career platforms, and online resources to gather information about available job roles, required skills, and employer expectations. This method is time-consuming, inconsistent, and often produces outdated or unreliable information, leaving graduates confused and underprepared when entering the job market. Additionally, the lack of a centralized and reliable source of job market information makes it difficult for graduates to identify the specific skills they need to develop in order to improve their employability (Stepanova et al., 2021).

Research has shown that employers in the computing industry consistently demand a combination of technical hard skills such as programming, data analysis, artificial intelligence, and database management, as well as soft skills such as communication, teamwork, and problem-

solving (Coetzee & Goede, 2024). However, many Computer Science graduates are unaware of these specific demands until they begin applying for jobs and facing rejection. Studies conducted across different regions including Saudi Arabia, Greece, South Africa, and the United States have all identified a consistent skills gap between what universities teach and what the industry requires (Karakolis et al., 2022; Alabdulkarim et al., 2023).

While some universities have begun to address this challenge by updating their curricula, most institutions including Augustine University Ilara-Epe still lack a dedicated system that provides graduates with structured, data-driven insights into the job market. Although graduates may have access to general career advice, there is no centralized platform that analyses real job market data and presents it in a clear and meaningful way to help Computer Science graduates understand employer demands and make informed career decisions.

The proposed Job Market Analysis System will include key modules such as:

- i. **Job Market Dashboard:** Allows graduates to view overall job market trends and insights in a simple and visual format.
- ii. **Skill Analysis Module:** Displays the most in-demand skills required by employers in the computing industry based on real job market data.
- iii. **Job Role Module:** Provides information on popular job roles available to Computer Science graduates along with their requirements.
- iv. **Search and Filter Module:** Allows graduates to search and filter job market information based on specific skills or job roles of interest.

By providing Computer Science undergraduates with access to accurate, data-driven job market insights, this system will reduce the time spent searching for career information, highlight skill gaps, and guide graduates in making better career decisions. Additionally, having a centralized platform will improve the overall awareness of job market demands among graduates and academic staff at Augustine University Ilara-Epe, ultimately contributing to improved graduate employability and better alignment between academic programmes and industry expectations.

Ultimately, the implementation of a Job Market Analysis System will offer a seamless, informative, and user-friendly approach to understanding the Computer Science job market, ensuring that graduates can prepare adequately for employment with minimal confusion and delay.

1.2 STATEMENT OF PROBLEM

The job market for computer science graduates is becoming more competitive, yet most graduates enter the work field without a clear grasp of what employers are looking for. Graduates have to manually explore many job listing websites and employment portals to get information that is sometimes inconsistent, obsolete, and fragmented, leading to bad career decisions. There is also a large skills gap between the CS curriculum and the skills that employers are looking for. Many graduates have strong theoretical knowledge but lack practical skills such as machine learning, data analysis, communication, and teamwork (Ibrahim et al., 2024; Stepanova et al., 2021).

This mismatch leads to rising unemployment of graduates coupled with an increasing need for computing specialists.

Additionally, no credible and data-based mechanism exists for academic institutions to track job market developments and align their curricula properly, resulting in leading universities continuing to generate graduates with abilities that are out of sync with the industry (Karakolis et al., 2022; Coetzee & Goede, 2024). Given these challenges, there is a need for an automated Job Market Analysis System for Computer Science graduates that collects and analyses job market data, identifies in-demand skills and roles, highlights skill gaps, and provides data-driven insights for graduates and institutions to improve employability and bridge the gap between academic training and industry expectations.

1.3 AIM AND OBJECTIVES OF THE STUDY

Aim

To design and implement a Job Market Analysis System that helps Computer Science graduates identify in-demand skills, understand employer requirements, and improve their chances of securing employment.

Objectives

The objectives of this computer software-based Job Market Analysis System for Computer Science Graduates include the following:

- i. To identify the challenges faced by Computer Science graduates in navigating the current job market.
- ii. To design a job market analysis system based on (i).
- iii. To implement the designed system.
- iv. To evaluate the system's performance.

1.4 RESEARCH QUESTIONS

In an effort to assist the design and implementation of the Job Market Analysis System for Computer Science graduates, the following research topics were formulated:

1. What are the most in-demand skills and job roles required by employers hiring Computer Science graduates in the current job market?
2. How can a web-based Job Market Analysis System improve the access of Computer Science graduates to reliable and data-driven job market insights?
3. To what extent is there a skills gap between the abilities obtained by Computer Science graduates in their academic study and the skills sought by employers in the computing industry?
4. How can data visualisation techniques be used to present complex job market information in a simple and easy-to-understand format for Computer Science graduates?
5. To what extent can an automated Job Market Analysis System reduce the time and effort spent by Computer Science graduates on manual job market research?

6. How can the proposed Job Market Analysis System help tertiary institutions such as Augustine University Ilara-Epe to connect their Computer Science syllabus with the current industry needs and employer expectations?

1.5 METHODOLOGY OVERVIEW

The methodology that will be used for the proposed system will be as follows:

- i. To identify the challenges a structured survey of existing job market systems will be done and key pain points will be identified from secondary sources.
- ii. The system will be designed using UML diagrams.
- iii. The system will be implemented using React.js, Chart.js and a pre-processed Kaggle dataset.
- iv. The system will be evaluated using ISO 9126 quality metrics: functionality, reliability, usability, and efficiency.

1.6 SCOPE OF THE STUDY

The project is designed for the management, staff, and Computer Science graduates of Augustine University Ilara-Epe to address the problems of job market navigation and career decision making. Specifically, the system focuses on analyzing trends in the labor market and requests of employers relevant to the graduates of the Computer Science department using a pre-collected dataset from Kaggle.

The system scope is to provide graduates with demand skill information, relevant employment roles and visual analytics based on the loaded dataset. The system will offer charts and graphs of

the most sought talents, popular job roles and current trends in the Computer Science job market in a straightforward and easy to comprehend manner.

This system is designed just to analyse the employment market of the Computer Science graduates and not other subjects of study. It won't be a job application portal or a direct link between graduates and companies. In addition, it does not require user login or database maintenance therefore keeping the system as simple and straightforward as feasible. It is an analytical and information tool that will help Computer Science graduates of Augustine University, Ilara-Epe make educated career choices and improve their employability based on current job market expectations.

1.7 SIGNIFICANCE OF THE STUDY

1. Improved Career Decision-Making for Graduates: One of the main advantages of this approach is that it offers Computer Science graduates of Augustine University Ilara-Epe precise and dependable data about the requirements of companies in the labour market. Graduates may obtain all essential job market facts in one location, instead of wasting time scrolling through a bunch of websites and platforms manually, to make better and more educated career

2. Bridging the Skill Gap: This system reveals the gap between the skills graduates have today and the talents companies actually want. The method also makes graduates aware of these gaps and makes them focus on getting the proper skills before entering the job market. This immediately addresses the identified gap between academic training and industrial expectations in the existing research (Ibrahim et al., 2024; Coetzee & Goede, 2024).

3. Benefit to Academic Institutions: The system also provides benefits to institutions and course coordinators with data-driven information about current employment market trends. The information can guide the academic staff of Augustine University Ilara-Epe in reviewing and updating their Computer Science curriculum to fit with the industry's demands and generate more employable graduates

4. Real-Time Job Market Insights: Graduates can view up-to-date information on the most in-demand talents, popular employment positions, and what employers are looking for, all presented in easy-to-understand charts and graphs. The real-time insight means that grads don't have to keep checking the job market manually to find out what's going on, and it ensures that they always get up-to-date and relevant professional information.

5. Contribution to Research: The study contributes to the expanding research on graduate employability and job market analysis. This project contributes to the field by developing a practical system to solve real problems faced by Computer Science graduates and shows how technology can be used to improve graduate employment outcomes.

6. Data-Driven Insights: The system provides meaningful visual analytics of the job market dataset for the graduates use in identifying trends, understanding employer needs, and making data-driven decisions. This data-driven strategy guarantees that the information provided is accurate, consistent and useful in improving the employability of Computer Science graduates.

In summary, the importance of establishing a Job Market Analysis System is to improve career decision-making, bridge the skill gap, guide curriculum development, provide timely job market information, support academic research, and provide data-driven information to Computer

Science graduates of Augustine University Ilara-Epe. These benefits result in a generation of Computer Science graduates that are better informed, prepared and employable.

1.8 OPERATIONAL DEFINITION OF TERMS

Job Market: The job market is the place where employers look for employees and graduates look for job opportunities. In the context of this study, it refers specifically to the computing and technology industry where Computer Science graduates seek employment after graduation.

Job Market Analysis: This is the gathering, processing and interpretation of information on the availability of jobs, employer requirements and skill demands in a particular sector. The system analyses the job market based on a pre-collected dataset, and provides Computer Science graduates with useful career insights.

Computer Science Graduate: A student who has successfully completed a bachelor's degree program in Computer Science or a related computing field and is preparing to enter the job market.

Skill Gap: A skill gap is the difference between the skills employers seek from job applicants and the skills graduates have after completing their academic programs. The method tries to identify such gaps to assist graduates enhance their employability.

Dataset: A dataset is a structured collection of related data used for analysis and processing. In this project, a pre-collected job market dataset obtained from Kaggle is used to provide

information on job listings, required skills, and employer demands relevant to Computer Science graduates.

System: A computer system is a complete and functional software setup designed to perform specific computing tasks. In this project, the system refers to the Job Market Analysis System built to help Computer Science graduates understand job market trends.

React.js: React.js is a JavaScript library used for building interactive and responsive user interfaces. In this project, React.js is used to develop the frontend dashboard where graduates can view job market trends, skill analyses, and job role information.

Chart.js: Chart.js is a JavaScript library used for creating visual representations of data such as bar charts, pie charts, and line graphs. In this project, Chart.js is used to display job market analytics in a simple and easy-to-understand visual format.

Dashboard: A dashboard is a visual display of important information organized in a simple and easy-to-read layout. In this system, the dashboard serves as the main interface where graduates can access job market insights and skill demand information.

Kaggle: Kaggle is a publicly available online platform that provides free datasets for research and analysis purposes. In this project, Kaggle is used as the source of the job market dataset that powers the analysis within the system.

Employability: Employability refers to the ability of a graduate to secure and maintain employment based on the skills, knowledge, and attributes they possess. This system aims to

improve the employability of Computer Science graduates by providing them with relevant job market information.

Job Role: A job role refers to a particular position or title in a company that involves specific duties and skill sets. Examples of job titles applicable to this project are Software Developer, Data Scientist, and Cyber security Analyst.

CHAPTER TWO

LITERATURE REVIEW

2.1 INTRODUCTION OF JOB MARKET ANALYSIS SYSTEMS

This chapter presents the conceptual and theoretical framework, along with a review of related and empirical literature relevant to the research topic. It critically examines existing literature on key definitions, concepts, study variables, and the factors influencing these variables as part of the conceptual review.

A Job Market Analysis System is a digital platform developed to simplify and automate the process of gathering, processing, and presenting employment market data exclusively for Computer Science graduates. Replacing the old manual way of searching through many job listing websites and career platforms, it boosts efficiency, minimises confusion, and accelerates the process of career decision-making for graduates. Additionally, it provides better insights into in-demand skills, popular job roles, and employer requirements, while facilitating seamless access to relevant job market information for both graduates and academic institutions, ensuring a more streamlined and informed approach to graduate employability.

The rapid growth of the technology industry has made it increasingly vital for Computer Science graduates to keep educated about current labour market trends and employer demands. As stated by Stepanova et al. (2021), many graduates enter the industry without a clear grasp of what employers are seeking for, resulting to high unemployment rates despite the growing demand for computing specialists. A Job Market Analysis System tackles this difficulty by offering graduates with a consolidated, data-driven platform that bridges the gap between academic training and industry expectations.

2.2 CONCEPTUAL REVIEW OF JOB MARKET ANALYSIS SYSTEMS

According to Wikipedia, Information Systems is an academic study of systems with a specific reference to information and the complementing networks of hardware and software that individuals and organisations use to gather, filter, process, create, and distribute data.

Alternatively, an information system is the interplay of people, processes, data, and technology (Singh, 2017). Information Technology is essential to the construction and operation of a Job Market Analysis System changing the old manual process of job market research into a fast, automated and efficient digital solution.

Through Information Technology, job market data is saved on organised data sets rather than spread across many platforms and physical resources. This makes it easy to access, evaluate and retrieve the data without the danger of inconsistency or obsolete information. As an internet-based system, it allows Computer Science graduates and academic staff to login from anywhere to get the insight of the job market, see the analyses of skill demand and explore the appropriate career openings easily. Information Technology also enhances the trustworthiness of job market information through the use of structured data processing methods that ensure that the insights offered to users are accurate and consistent.

A Job Market Analysis System needs some critical technologies to perform properly. Easy to access and manage job market information is found in datasets from publicly available sources like Kaggle, with job listing records, required skills, and company expectations. The user interface is built using web design tools including HTML, CSS, and Javascript, and backend languages like Node.js for processing and data administration. The dynamic frontend dashboard

is developed using React.js to allow the graduates to study the trends of the job market and skill needs in a visual and user-friendly way.

The system uses data presentation tools such as Chart.js to provide complex job market data in the form of easy-to-understand charts and graphs such as bar charts of the most in-demand talents and pie charts of the distribution of popular employment roles. The visual representation of job market data allows graduates to immediately grasp and act on the system's insights without needing any technical expertise.

Furthermore, automation removes the necessity for graduates to individually look through numerous job listing websites, as it immediately processes and delivers pertinent job market information on a single, consolidated platform, thereby dramatically decreasing the time and effort needed for career exploration. Notification and filtering tools allow graduates to search and filter job market information based on specific talents or job roles of interest, offering a transparent and well-organised career advising experience.

The technologies provide faster, more accurate and convenient access and use of relevant job market information for Computer Science graduates and academic administrators at Augustine University Ilara-Epe which leads to better graduate employability and career decision-making, making the job market analysis easier, taken together.

2.3 THEORETICAL CONCEPTS OF JOB MARKET ANALYSIS SYSTEMS

According to Kombo and Tromp (2006) in *Proposal and Thesis Writing: An Introduction*, a theoretical concept is "a set of interrelated principles or ideas that provide a systematic view of a phenomenon by specifying relations among variables, with the purpose of explaining and

predicting the phenomenon." The working principles of Job Market Analysis Systems are explained here. The usefulness of the system, how it enhances the job market analysis process, and the reasons why graduates and institutions embrace or reject it are all explained by theories. A few significant theories are as follows:

Information System Theory: Information System Theory covers the technology required for the efficient processing and management of information. This theory supports the premise that within the framework of a Job Market Analysis System, digital platforms may collect, store, process and show enormous amounts of job market data in a structured and understandable way. The idea postulates that an efficient information system must be dependable, accurate and accessible for its users (Laudon & Laudon, 2016). This is even more so in this project as the system is designed to offer Computer Science graduates accurate and current job market information in one centralised location therefore reducing the need for inefficient manual job market research.

Automation Theory: Automation Theory explains how procedures that were formerly completed manually can now be completed by computers, saving time and reducing errors. In relation to this project, automation theory supports the transition from manual job market research, where graduates individually search through multiple platforms, to an automated system that processes and displays relevant job market data instantly. This theory is particularly significant as it justifies the need for a Job Market Analysis System that automates the collection and presentation of job market insights, reducing the time and effort required by graduates to understand employer demands and industry trends (Chakraborty et al., 2019).

Technology Acceptance Model (TAM): The Technology Acceptance Model (TAM), first presented by Davis (1989), and it describes the adoption of new technology by individuals based on two main constructs: perceived usefulness and perceived ease of use. Perceived usefulness is the degree to which a user believes that using a specific system will enhance his/her performance (Davis, 1989), whereas perceived ease of use is the degree to which a user believes that utilising a particular system will be devoid of effort (Davis, 1989). Within the framework of this research, TAM helps to explain why Computer Science graduates and academic staff at Augustine University Ilara-Epe would embrace and use the Job Market Analysis System. If graduates regard the system as valuable in helping them grasp job market trends and straightforward enough to access, they are more inclined to adopt and profit from it consistently.

Data Analytics Theory: Data Analytics Theory discusses how the raw data is converted in to useful information via the process of collecting, processing and visualisation. This theory is a core part of the Job Market Analysis System development, as the system itself is based on assessing a pre-collected job market dataset to provide visual insights such as charts and graphs highlighting in-demand skills and popular job roles. Coetzee and Goede (2024) state that colleges and graduates need data-driven ways to understanding job market demands to close the skill gap and boost graduate employability. This way, the theory gives a robust background for the analytic functions of the proposed system.

2.4 TRADITIONAL METHOD OF JOB MARKET ANALYSIS

The traditional method of job market analysis, also known as the manual approach, relies on physical job searching, paper-based career resources, and unstructured information gathering techniques (Ben et al., 2015; Idachaba et al., 2015). Almost all Computer Science graduates

engage in some form of job market research before entering the workforce, and these graduates can use alternative methods of gathering career information depending on the available resources, technology, and personal choices (Zuhaib, 2013).

Traditionally, employment market analysis has been done by graduates manually browsing through various websites, newspapers, career fairs and word of mouth to obtain information on job responsibilities and requirements by different employers. This is a tedious and unorganised process that can take several weeks to complete and is unpleasant and has negative impacts on both the graduates and the academic institutions involved (Jonathan et al., 2019). This will hurt their productivity and delay their career preparation process as graduates will have to explore independently through various platforms and resources to gain a good image of the job market.

Moreover, conventional techniques of data collection are frequently inconsistent, obsolete, and incomplete. At times, the relevant career information may not be available and graduates will have to visit the same platforms multiple times before they can find useful and reliable job market data (Adamu, 2022). As Stepanova et al. (2021) explain, many graduates enter the workforce without a clear grasp of what employers are looking for, mainly because the traditional method to job market research does not offer structured or comprehensive career insights.

The internet is now available and need for speed and accuracy in career decision making is growing. As such educational institutions and graduates are now considering web-based solutions for job market research and career guidance (Techie-Menson et al., 2022; Techie-

Menson & Nyagorme, 2021). The disadvantages of the traditional method of analysing the job market are:

Time-consuming: Graduates spend excessive amounts of time manually searching through multiple platforms to gather job market information that could otherwise be presented in a single centralized system.

Inconsistent Information: Information gathered manually from different sources is often inconsistent, making it difficult for graduates to make reliable career decisions.

Lack of Data Analysis: The traditional method does not provide structured analysis of job market trends, skill demands, or employer requirements, leaving graduates without the data-driven insights they need to improve their employability.

Limited Accessibility: Career information gathered through traditional methods is not always easily accessible to all graduates, particularly those in developing countries where career guidance resources are limited (Coetzee & Goede, 2024).

No Real-Time Updates: Traditional methods do not provide real-time updates on changing job market trends, meaning graduates may rely on outdated information when making important career decisions.

Some institutions have begun using automated job market analysis systems as a means of addressing the aforementioned problems with the manual procedure, recognizing that a data-driven and centralized approach to job market analysis is essential for improving graduate

employability and bridging the skill gap between academic training and industry expectations (Karakolis et al., 2022).

2.5 AUTOMATED METHOD OF JOB MARKET ANALYSIS

The automated Job Market Analysis System is a career research tool providing Computer Science graduates with fast, efficient and transparent access to job market information. The automated solution still keeps the key job market analysis methods but accelerates the procedure, analyses and presents data automatically and interactively through visual dashboards and infographics. In addition, the automated system saves a lot of time, is user friendly, speeds up the processing of information and does not require graduates to manually search for multiple sites with job listings and career resources.

Because these web-based solutions leverage the internet, any graduate with an internet connection can get employment market insights at any time and from any location. This is particularly advantageous for Computer Science graduates from Augustine University Ilara-Epe who may not have easy access to structured career assistance resources. As mentioned by Karakolis et al. (2022), automated job market analysis solutions that employ data analytics and online technologies have shown to be highly effective in helping graduates find in-demand talents and align their career preparation with current industry expectations

Automated job market analysis systems offer several advantages over traditional manual methods, including:

Speed and Efficiency: Automated systems process and display large amounts of job market data almost instantly, saving graduates the time and effort required to manually gather and interpret career information from multiple sources.

Accuracy and Consistency: Unlike manual research methods, automated systems rely on structured datasets and algorithms to present consistent and reliable job market insights, reducing the risk of outdated or contradictory information.

Accessibility: Web-based automated systems can be accessed from any device with an internet connection, making job market information readily available to all graduates regardless of their location.

Data Visualisation: Automated systems present complex job market data in the form of easy-to-understand charts and graphs, enabling graduates to quickly identify trends, skill demands, and popular job roles without requiring technical expertise (Coetzee & Goede, 2024).

Real-Time Insights: Automated systems can provide up-to-date job market information, ensuring that graduates always have access to current and relevant career data when making important employment decisions.

It is important to keep in mind that these automated and web-based solutions still have significant shortcomings, which include inaccessible user interfaces, inadequate information coverage, and a failure to prioritize processes relevant to specific fields of study (Jonathan et al., 2019). For instance, many existing job market analysis tools are designed for a general audience and do not specifically cater to the needs of Computer Science graduates. This limitation further justifies the need for the proposed Job Market Analysis System, which is specifically tailored to

address the career guidance needs of Computer Science graduates at Augustine University Ilara-Epe.

2.6 JOB MARKET ANALYSIS SYSTEM AS A SOFT SYSTEM

A soft system is one which puts the emphasis on human interaction, decision making and flexibility rather than on strict automation and technology. A soft system differs from fully automated systems operating on predefined rules in that it is based on the fact that some processes require human judgement, adaptability and problem solving to deal with complex or exceptional cases.

In a Job Market Analysis System for Computer Science graduates, the soft system method guarantees that the job market analysis process is not a rigid, automated workflow but also allows human intervention where needed. The system automates data processing, skill analysis and visualisation of job market trends, but there are situations that require administrative decisions and human judgement which technology alone cannot deliver. For example,

1. Exceptional Cases: If the job market dataset contains missing, wrong, or outdated information, the system may highlight it, but an administrator must personally review and rectify the issue rather than rely entirely on automation. This ensures that the insights offered to graduates are correct and dependable at all times.

2. Curriculum Decision-Making: While the system can identify skill gaps and in-demand job roles based on data analysis, academic staff at Augustine University Ilara-Epe must use their professional judgment to decide how to incorporate these insights into curriculum updates and

academic programme adjustments. Technology alone cannot make these complex educational decisions.

3. Flexible Interpretation of Data: Some job trends demand an interpretation in context that an automated system cannot supply. A transient industrial fad is an example of this, where it may create a momentary need for a skill rather than a permanent change in the marketplace. Such patterns need to be interpreted by human experts in a wider context and to give relevant career advice to graduates.

4. Communication and Support: Unlike a fully automated system, a soft system integrates direct communication between graduates and academic advisors or career guidance officers, ensuring that graduates can seek clarification, get personalised feedback, or request assistance in interpreting the job market insights provided by the system. This human element is essential for ensuring that graduates fully benefit from the information presented.

5. Personalised Career Guidance: Every graduate has unique strengths, weaknesses, and career goals that a purely automated system cannot fully account for. The soft system approach allows career advisors to use the insights generated by the Job Market Analysis System as a foundation for providing personalised career guidance tailored to the individual needs of each graduate.

The Soft Systems Methodology (SSM) by Checkland (1981) is a well-known approach used in creating systems that integrate both technology and human decision-making. It helps universities construct a balanced job market analysis system where automation performs mundane duties such as data processing and visualisation, while staff members and career advisors intervene when human judgment is needed. As observed by Karakolis et al. (2022),

effective job market analysis systems must find a balance between automated data processing and human decision-making to guarantee that the insights supplied are both accurate and realistically valuable for graduates and institutions.

By incorporating both automation and human flexibility, a Job Market Analysis System using a soft system approach ensures efficiency, fairness, and a smooth career guidance experience for Computer Science graduates at Augustine University Ilara-Epe, ultimately contributing to improved graduate employability and better alignment between academic training and industry expectations.

2.7 JOB MARKET ANALYSIS SYSTEM AS A DYNAMIC SYSTEM

A dynamic system is one that changes and responds to changes throughout time . This implies that in a Job Market Analysis System for Computer Science grads, the system is able to:

Know when fresh job market data available, update job market insights & skills demand information.

Automatically adapt and respond to changes in employer demands, such as the increased need for artificial intelligence and data science capabilities described by Coetzee and Goede (2024).

Adapt and grow with the computing business as it continues to advance to new job functions, new technology, and new features.

Link up with other platforms and data sources to make sure the job market information shown to graduates is up-to-date and appropriate.

Sterman (2000) states, “Dynamic systems change over time as a function of their internal structure and the external environment, requiring ongoing adaptation in order for them to stay efficient and effective.” The same is true for a Job Market Analysis System, which needs to be efficient, flexible, and sensitive to the career guidance needs of Computer Science graduates and the constantly changing needs of the technology industry.

The fast rate of evolution of the computing industry makes the constantly changing nature of the Job Market Analysis System very vital. As stated by Ibrahim et al. (2024), the tech business has ever-changing demands from employers, therefore any tool that analyses the employment market needs to be flexible and updated periodically to keep up with the latest industry developments and employer expectations.

2.8 REVIEW OF RELATED WORKS

Several studies have been conducted on job market analysis systems to address the challenges faced by Computer Science graduates in navigating the job market and understanding employer demands. These works concentrate on automating the gathering of job market data, identifying skill gaps and offering data-driven career support to graduates and academic institutions.

Author(s) & Year	Study Focus	Methodology	Key Findings	Limitations	Relevance to This Study
Ibrahim et al. (2024)	Curriculum development and employment	Machine learning classifiers applied to a dataset of	Identified significant skill gaps between graduate competencies	Did not develop an interactive system for graduates to access	Highlights the skill gap problem this system aims to

Author(s) & Year	Study Focus	Methodology	Key Findings	Limitations	Relevance to This Study
	trends for CS graduates in Saudi Arabia and Malaysia	3,831 CS graduates	and employer demands; technical skills such as programming and data analysis were most in demand	findings; focused on academic analysis only	address through visual job market insights for CS graduates
Karakolis et al. (2022)	Bridging the gap between technological education and job market requirements in Greece	Web scraping of over 1,500 job postings from Greek job portals; data analytics and decision support	Provided data-driven insights on in-demand skills and roles; showed mismatch between university curricula and industry needs	Required advanced technical expertise to operate; not accessible to average graduates without a technical background	Demonstrates the value of data-driven systems; informs the need for a simpler, graduate-friendly interface in this project
Alabdulkarim et al. (2023)	Career paths of CS graduates across five computing majors in Saudi Arabia	Analysis of 2,055 LinkedIn profiles across Computer Science, Software Engineering, IT, IS, and CE	Software Engineering and Computer Science graduates had the widest range of job opportunities; skill diversity improved employability	Limited to Saudi Arabia; no centralized platform provided for graduates to independently explore the findings	Supports the need for a platform where graduates can explore job roles and skill demands relevant to their specific major
Stepanova et al. (2021)	Employer perspectives on hiring CS graduates in the United States	Survey of 218 recruiters hiring CS graduates; quantitative analysis of	Software developer was the most in-demand role (63% of responses); work experience, communication, and	Survey-based only; findings not translated into a digital tool graduates can use for career guidance	Provides employer-side data that informed the job role and skill

Author(s) & Year	Study Focus	Methodology	Key Findings	Limitations	Relevance to This Study
		employer priorities	problem-solving ranked above GPA		demand features of this system
Chakraborty et al. (2019)	Graduate performance analysis and ranking for CS and Engineering graduates in Bangladesh	Decision tree algorithms and Pearson Correlation applied to data from ten universities	Generated individual graduate rankings based on academic and skill factors; identified employability predictors	Focused on graduate ranking rather than job market trends; did not address employer skill demands or industry requirements	Reinforces the importance of data-driven analysis in graduate employability; informs the analytical approach of this project
Coetzee & Goede (2024)	Employability competencies needed by data analytics graduates in South Africa	Three-step pluralistic method: qualitative analysis with Atlas TI and quantitative analysis of 156 online job listings	Identified technical, analytical, and soft skills as critical for employability; growing demand for AI and data science skills	Focused exclusively on data analytics graduates; does not cover the broader Computer Science job market	Highlights emerging skill trends relevant to CS graduates; supports the skill gap analysis feature of this system

Table 2.1: Review of Related Works

Ibrahim et al. (2024) conducted a study that mapped graduate skills to market demands through a holistic examination of curriculum development and employment trends. Using a dataset of

3,831 Computer Science graduates from universities in Saudi Arabia and Malaysia, the study applied machine learning classifiers including Support Vector Machine and Artificial Neural Networks to analyse the relationship between graduate skills and employer requirements. The study identified the top five in-demand skills as project management, Java, statistics, deep learning, and machine learning. While the study provided valuable insights into the skill gap between graduates and employers, it was limited to a specific geographical region and did not provide a practical system that graduates could directly interact with to access job market insights.

Karakolis et al. (2022) propose a holistic approach that employs data analytics and decision support services to bridge the gap between technological education and the needs of the job market. The system included a web scraping module to collect more than 1,500 job postings from Greek job portals; a skill extraction module based on Natural Language Processing and ESCO classification; and a course recommender based on clustering algorithms and association rule mining to detect missing skills in existing curricula. The system also integrated multi-criteria decision support techniques such as TOPSIS and PROMETHEE II to assist students in making educated course selection decisions. The system was incredibly detailed, but was only reviewed by 31 users and was limited to the Greek employment market, making it less suitable to graduates in other locations.

Alabdulkarim et al. (2023) investigated the job market perspective of computing undergraduate majors with a dataset of 2,055 LinkedIn profiles from Saudi Arabia. The study analysed the careers of graduates in five computer fields: Computer Science, Software Engineering, Information Technology, Information Systems, and Computer Engineering. Key findings showed

that the most common initial career for graduates was software development, with 90% of graduates being employed within two years after graduation. The report also found that need for business analysis, artificial intelligence and cybersecurity skills is expanding, while traditional roles like system administration and technical assistance are falling. However, the study was limited to LinkedIn profiles of one country and did not give a practical platform for graduates to obtain these insights directly.

To understand what firms look for when hiring, **Stepanova et al. (2021)** surveyed 218 recruiters hiring Computer Science graduates in the United States to understand what employers look for when hiring. The study found that software developer was the most in-demand position, accounting for 63% of all responses, and that recruiters prioritised work experience, GPA, and personal projects when reviewing graduate resumes. The study also highlighted a significant gap between CS undergraduate education and employer expectations, particularly in soft skills such as communication, teamwork, and problem-solving. While the study provided valuable insights from the employer perspective, it was limited to recruiters from the state of Texas and did not address the needs of graduates in developing countries.

Chakraborty et al. (2019) proposed a decision tree method and Pearson Correlation based system for analysing the Computer Science and Engineering graduates of different universities of Bangladesh. The system is trained on data from 10 universities and develops individual rankings for graduates based on CGPA, and programming skills and the time taken to get the first job. The system also provided expected salaries and estimated time to get a job for graduates. The system demonstrated a practical method to analyse graduate employment data, but was limited to

Bangladesh and focused mainly on university rankings, instead of offering a thorough insight into the job market for graduates.

Coetzee and Goede (2024) performed a study to analyse the employability competencies required of data analytics graduates based on 156 online job advertisements in South Africa. The study employed a three-step pluralistic methodology, integrating qualitative analysis with Atlas TI and quantitative analysis with R, and identified SQL, Python and Excel as the most in-demand data analytics software skills. Artificial intelligence was the most requested hard skill, showing up in 82% of job advertisements. The study also found teamwork, business knowledge and management skills as the top soft skills that employers want. While the study gave a comprehensive summary of the skills South African employers are seeking, it was limited to the data analytics field and did not focus specifically on the wider Computer Science job market.

Summary of Related Works

All the reviewed studies above offer valuable insights into the relationship between Computer Science graduate skills and job market needs. However, a common limitation across these studies is that they either focus on specific geographical regions, require advanced technical expertise to implement, or do not provide a practical and user-friendly platform that graduates can directly interact with to access job market insights.

In view of the above limitation, the proposed Job Market Analysis System is intended to provide a simple, centralised and interactive platform for Computer Science graduates at Augustine University Ilara-Epe to easily access data-driven job market insights, identify in-demand skills

and make informed career decisions based on current industry demands. The system was developed with HTML, CSS, JavaScript and Chart.js. As opposed to the studies reviewed, the system also includes primary data from Nigerian employers and graduates, giving it a localised perspective that is largely missing from the existing literature.

COMPARISON AND LIMITATIONS OF EXISTING SYSTEMS

While successful job market analysis systems have been developed in some researches, they are still limited in some aspects. Most existing systems focus on data collection and analysis for specific regions or fields, but many lack practical and user-friendly interfaces that graduates can directly interact with to access job market insights. For example:

Ibrahim et al. (2024) provided valuable insights into skill gaps using machine learning classifiers but did not develop a practical interactive system that graduates could use to access job market information directly.

Karakolis et al. (2022) developed a comprehensive system with multiple components but required advanced technical expertise to operate, making it less accessible to average graduates without a technical background.

Alabdulkarim et al. (2023) analysed LinkedIn profiles to identify job market trends but was limited to a single country and did not provide a centralized platform for graduates to explore these insights independently.

Stepanova et al. (2021) gathered valuable employer perspectives through surveys but did not translate these findings into a practical digital tool that graduates could use for career guidance.

Coetzee and Goede (2024) identified key employability competencies through job listing analysis but focused exclusively on the data analytics field, leaving out the broader Computer Science job market.

From this comparison, it is evident that while existing studies attempt to solve job market analysis challenges, they still have gaps in accessibility, practical usability, and specificity that need to be addressed.

HOW THIS PROJECT IMPROVES ON EXISTING SYSTEMS

This research proposes a simple and practical Job Market Analysis System that addresses the shortcomings of previous works. Unlike earlier studies, this project integrates the following improvements:

1. **User-Friendly Interface:** A simplified and intuitive dashboard built with HTML, CSS, and JavaScript will make it easy for Computer Science graduates at Augustine University Ilara-Epe to navigate and access job market insights without requiring any technical expertise.
2. **Visual Data Representation:** Unlike existing studies that present findings in academic reports and research papers, this system uses Chart.js to display job market insights in the

form of easy-to-understand charts and graphs, making the information more accessible and actionable for graduates.

3. **Centralized Job Market Information:** Rather than requiring graduates to search through multiple platforms and resources, this system provides a single centralized platform where all relevant job market data, skill demands, and job role information are available in one place.
4. **Specifically Tailored for Computer Science Graduates:** Unlike existing systems that focus on general job markets or specific fields such as data analytics, this system is specifically designed to address the career guidance needs of Computer Science graduates, ensuring that the insights provided are directly relevant to their field of study.
5. **Localized Primary Data:** Unlike existing studies that rely solely on secondary or global datasets, this system incorporates primary data collected directly from Nigerian employers and Computer Science graduates, providing insights that reflect the realities of the local job market.

By incorporating these features, this system offers a more accessible, practical, and user-friendly approach to job market analysis, significantly improving upon existing research and providing Computer Science graduates at Augustine University Ilara-Epe with the tools they need to make informed career decisions.

CHAPTER THREE

METHODOLOGY

3.1 INTRODUCTION

This chapter explains the methods and tools used in developing the Job Market Analysis System for Computer Science graduates. It covers the research design, system development approach, data collection methods, and system analysis. Each section is structured to directly address the objectives of the study as stated in Chapter One.

3.2 METHODOLOGY OF THE STUDY

To achieve the objectives of the system, the methodology to develop the system will be as follows:

3.3 DATA COLLECTION METHODOLOGY

To achieve **Objective (i)**, data was obtained through primary and secondary sources.

3.3 DATA COLLECTION METHODOLOGY

To achieve Objective (i), data was obtained through primary and secondary sources.

3.3.1 PRIMARY SOURCE

Primary data was collected through two structured questionnaires distributed via Google Forms; one administered to Computer Science graduates and final year students of Augustine University Ilara-Epe, and the other administered to Nigerian technology employers. A total of 11 graduate responses and 20 employer responses were received.

The graduate questionnaire gathered information on the challenges respondents face when searching for job market information, the skills they currently possess, their salary expectations, and their preferred work arrangement. The employer questionnaire gathered information on the industries represented, the roles and skills employers prioritize when hiring fresh graduates, typical starting salaries, and the challenges employers face when recruiting Computer Science graduates.

3.3.2 SECONDARY SOURCE

Secondary data was obtained from the Kaggle (AI and Data Science Job Market) Dataset also served as a key secondary source, providing 10,345 real job listing records spanning 2020 to 2026 across seven countries. This dataset supplied the job role, skill demand, salary, and geographic data used to power the system.

3.4 ANALYSIS OF THE EXISTING SYSTEM

To achieve **Objective (i)**, the existing system was examined to identify the specific challenges it presents to Computer Science graduates at Augustine University Ilara-Epe.

The existing method of job market research is largely manual and unstructured. Graduates independently search through multiple job listing websites and career platforms to gather information that is often inconsistent, outdated, and unreliable, leaving them confused and underprepared when entering the job market.

3.4.1 PROBLEMS OF THE EXISTING SYSTEM

Graduates spend excessive time manually searching through multiple platforms for job market information.

Information gathered is often inconsistent and outdated, making reliable career decisions difficult.

There is no centralized platform providing structured job market insights specific to CS graduates.

Academic staff lack a reliable tool to monitor job market trends and update the curriculum accordingly.

3.4.2 DESCRIPTION AND JUSTIFICATION FOR THE PROPOSED SYSTEM

To achieve Objective (ii), the proposed Job Market Analysis System will address all limitations of the existing manual approach by providing graduates with a centralized, automated, and user-friendly platform.

The system offers a dashboard for job market overviews, visual charts powered by Chart.js for skill analysis, a search and filter feature for personalized exploration, and a simple HTML, CSS, and JavaScript interface accessible to all graduates regardless of technical expertise. The system is further strengthened by a dedicated Nigerian Insights module, built from primary survey data, which compares the skills Nigerian employers demand against the skills graduates currently possess.

3.4.3 REQUIREMENTS ANALYSIS

To achieve Objective (ii), the system requirements are captured in two categories:

Functional Requirements

1. The system should display a dashboard showing job market trends, in-demand skills, and popular job roles.
2. The system should provide visual charts showing the most sought-after technical skills.
3. The system should display popular CS job roles along with their skill requirements and detailed statistics for each role.
4. The system should allow graduates to search and filter job market information by role, skill, country, experience level, and work arrangement.
5. The system should provide summary insights highlighting key job market and skill gap information, including a comparison between Nigerian employer demand and graduate skill supply.
6. The system should retrieve live job listings from an external job listing source.

Non-Functional Requirements

1. Usability: The system should have a clear and easy-to-navigate interface accessible to all graduates without technical expertise.
2. Performance: Job market data and charts should load quickly to ensure a smooth user experience.
3. Reliability: The system should be consistently available with a good fault tolerance mechanism.

4. Scalability: The system should accommodate multiple users simultaneously without affecting performance.
5. Accuracy: Job market information displayed must be accurate and based on reliable data sources.
6. Maintainability: The system should be easy to update as the computing industry continues to evolve.

3.5 SYSTEM DESIGN

To achieve **Objective (ii)**, the system was designed using UML (Unified Modelling Language) diagrams and a clearly defined system architecture.

The design phase translates the requirements identified in Section 3.4.4 into a structured blueprint that guided the implementation of the proposed Job Market Analysis System. The system design covers the system architecture, use case diagram, system flowchart, and entity relationship diagram.

3.5.1 SYSTEM ARCHITECTURE

The system architecture describes the overall structure of the proposed Job Market Analysis System. The system adopts a client-side architecture consisting of three layers:

User Interface Layer: Built with HTML5, CSS3 and JavaScript, this layer presents all job market information to the user through an interactive and easy-to-navigate web interface. It handles all user interactions including navigation, search, and filtering.

Visualisation Layer: Powered by Chart.js, this layer converts raw job market data into visual charts and graphs including bar charts, pie charts, and horizontal skill bars for easy interpretation by graduates.

Data Layer: The pre-processed Kaggle dataset stored as a CSV file serves as the primary data source. It is loaded directly into the React application where JavaScript functions process and prepare it for display across the different modules of the system.

3.5.2 USE CASE DIAGRAM

The use case diagram illustrates the interaction between the system user, namely the CS Graduate, and the functions available to them within the Job Market Analysis System. The CS Graduate can view the dashboard, explore skill analysis, browse job roles, view detailed statistics for a specific role, use the search and filter feature, review salary trends, explore geographic job distribution, and access the Nigerian Insights module.

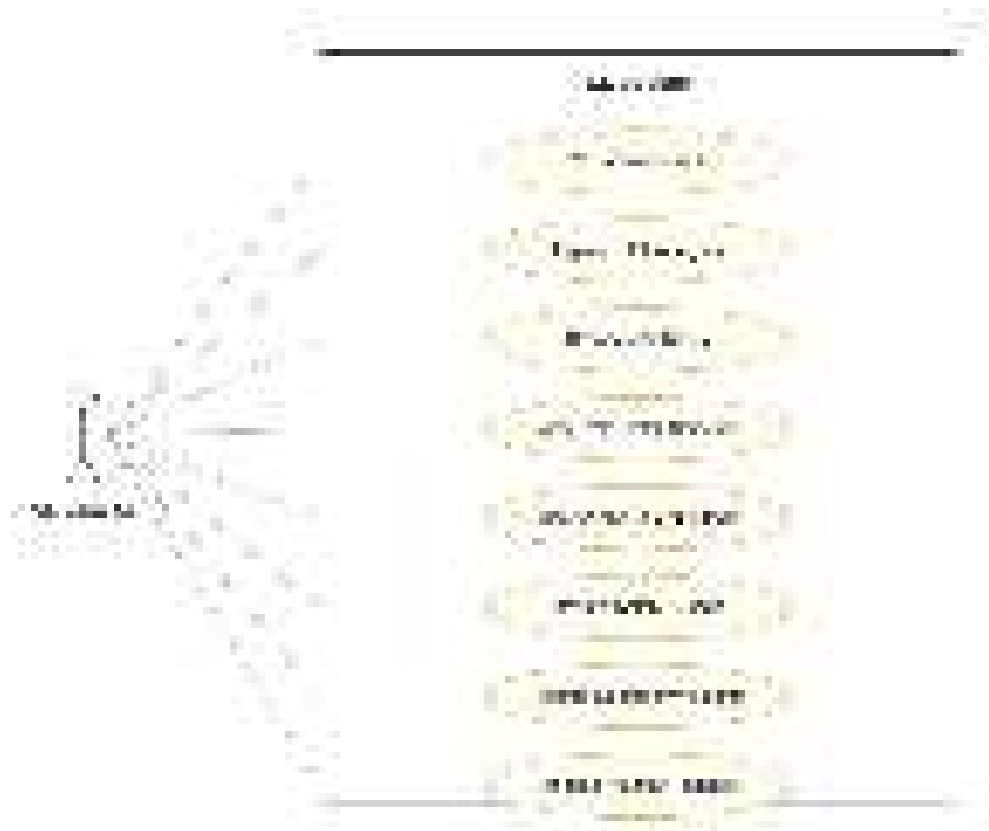


Figure 1 Use Case Diagram

3.5.3 SYSTEM FLOWCHART

The system flowchart represents the logical flow of the Job Market Analysis System from the point where the user opens the platform to the point where they access the job market insights they need. The flowchart clearly illustrates the decision points, processes, and outcomes involved in using the system.

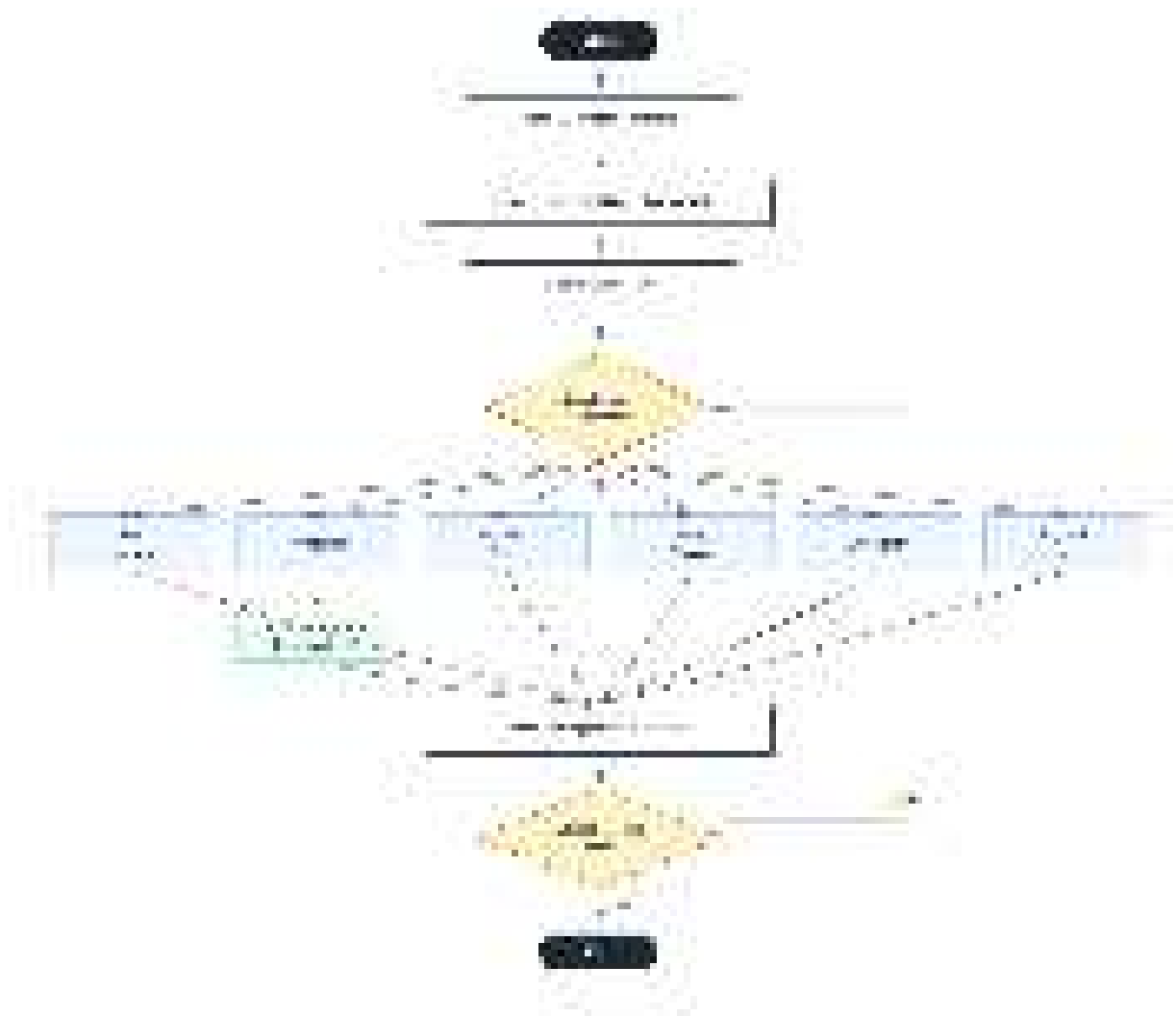


Figure 2 System Flowchart

3.5.4 ENTITY RELATIONSHIP DIAGRAM (ERD)

The Entity Relationship Diagram shows the relationship between the key entities of the Job Market Analysis System, namely Job Listings, Employer Responses, Graduate Responses, and Live Job Cache. The ERD provides a clear representation of how these entities relate to one another within the system and forms the basis for the data structure used in the application.



Figure 3 Entity Relational Diagram

3.5.5 SYSTEM ARCHITECTURE DIAGRAM

The Job Market Analysis System is built on a three-layer architecture comprising the Frontend Presentation Layer, the Backend Logic Layer, and the Database Layer. This separation of concerns ensures that each layer can be developed, tested, and maintained independently, while still working together to deliver job market insights to the user.

The **Frontend Presentation Layer** consists of eight HTML pages; Dashboard, Skill Analysis, Job Roles, Role Detail, Search and Filter, Salary Trends, Geography, and Nigerian Insights built using HTML5, CSS3, and vanilla JavaScript, with Chart.js used to render all data visualisations.

When a user navigates to any page, the corresponding JavaScript file sends an HTTP request to the backend to retrieve the data needed for that page.

The **Backend Logic Layer** is built with Node.js and Express.js, and exposes twelve REST API endpoints (including overview, skills, roles, salary, geography, search, live, and the four Nigerian Insights endpoints under api/nigerian). Each endpoint receives the frontend's HTTP request, queries the appropriate table in the database, processes the result, and returns it as a JSON response. The backend also handles communication with the Adzuna API to retrieve live Nigerian job listings on demand.

The **Database Layer** uses SQLite, accessed through the better-sqlite3 library, to store all data used by the system. As shown in the Entity Relationship Diagram, the database consists of four independent tables: jobs, which stores the 10,345-record Kaggle dataset seeded into the system on first run; employer responses and graduate responses, which store the primary data collected from the Nigerian employer and graduate questionnaires; and live_jobs_cache, which stores the most recent set of job listings fetched from the Adzuna API.

The diagram below illustrates the flow of data between the three layers. When a user interacts with the system, a request travels from the Frontend Layer to the Backend Layer, which queries the Database Layer, processes the returned data, and sends a JSON response back to the Frontend Layer for display. This request-response cycle repeats independently for each page the user visits, ensuring that only the data required for the active page is loaded at any given time.

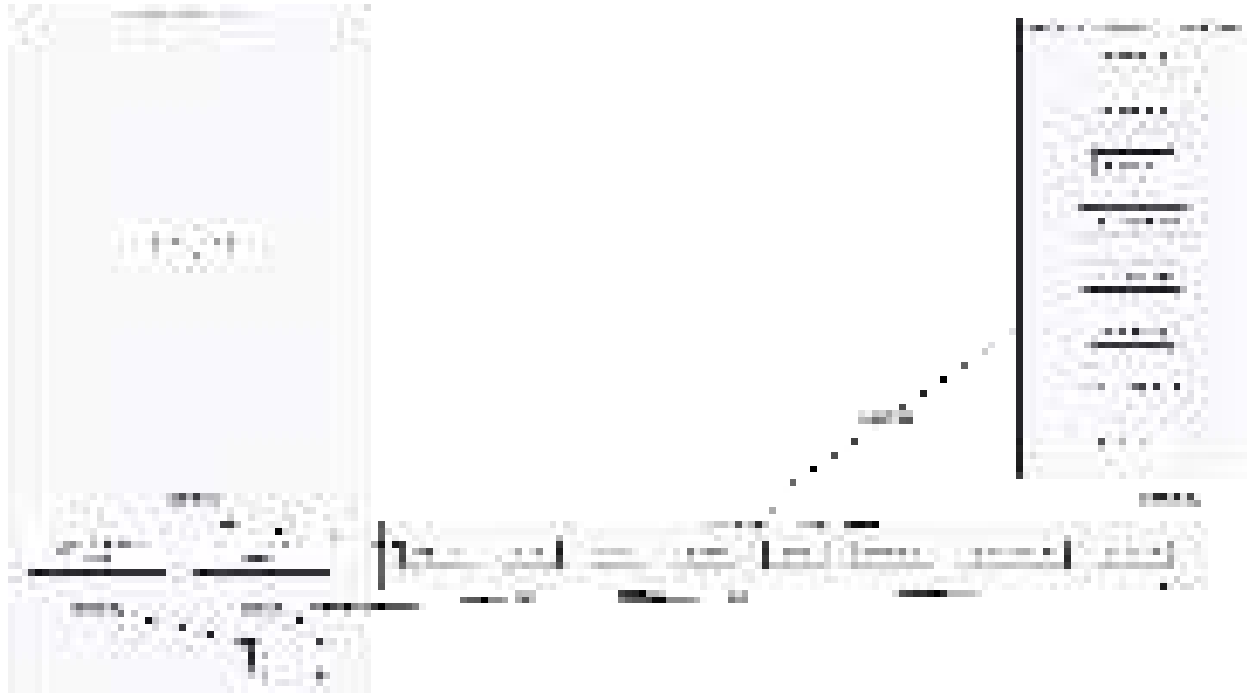


Figure 4 System Architecture Diagram

3.5.6 INTERFACE LAYOUTS

The Job Market Analysis System is eight pages long and each page is designed to present a particular type of job market information in a clear and accessible way. All pages share the same layout structure: a fixed top bar for system name and dataset information, a fixed left sidebar for navigation, and a scrollable main content area for visualisations of data and insights.

The Dashboard page is the entry point of the system. It includes four summary metric cards: total records, average salary, high urgency percentage, and top role, followed by a role distribution doughnut chart, an average salary bar chart, a job posting trend line chart from 2020 to 2026, a skill demand overview section and a work type distribution chart.

On the Skill Analysis page, you can see how much demand there is for each of the 5 skills we track — Python, SQL, Machine Learning, Deep Learning, and Cloud — across all 10,345 listings. It has individual skill metric cards, overall skill demand bars, a grouped bar chart for skill demand by job role and a second grouped bar chart for demand by experience level.

The Job Roles page shows a structured table with all six job roles in the dataset, with columns for average salary, total listings, top skill, and hiring urgency. Each row is clickable and will take the user to Role Detail page for that specific role.

Role Detail page – A dedicated six-year analysis for any role selected from the Job Roles page. The report contains a bar chart of listings per year, a salary trend line chart, a skill demand breakdown, a hiring urgency stacked bar chart, a work type split chart and an experience level distribution doughnut chart with written insight summary at the bottom of the page.

The Search and Filter page allows the user to filter across all 10,345 job listings based on a combination of role, country, experience level and work type criteria. Results are shown as cards with the job title, location, experience level, salary and skills required. There is also a Live Nigerian Jobs tab on the page which fetches and displays current Nigerian job listings from Adzuna API.

The Salary Trends page analyses salaries across all dimensions in the dataset – by job role, by experience level per role, by year, and by country. The page uses a combination of horizontal bar charts and line charts, and all salary values are in USD.

The Geography page shows the distribution of job listings and average salaries across the 7 countries in the dataset. This includes a bar chart for listing count and salary by country, a work type distribution chart and a breakdown of top industries.

The Nigerian Insights page displays the core data collected from the Nigerian employer and graduate surveys. It includes a skill gap analysis chart comparing skills that employers want vs skills that graduates have, a salary expectations vs reality comparison chart with values in Nigerian Naira, charts for target roles, employer industries, and work preferences, and a numbered list of open-text responses from both employers and graduates.

3.6 DEVELOPMENT TOOLS

To achieve objective (iii), the following tools and technologies were used in the design and implementation of the Job Market Analysis System.

System Layer	Technology Selected
Frontend Presentation	HTML5 + CSS3 + Vanilla JavaScript
Visual Analytics	Chart.js (CDN Integrated)
Backend Engine	Node.js + Express.js
Database Engine	SQLite (better-sqlite3)
Live Data Integration	Adzuna API (Nigerian job listings)
Development Environment	Visual Studio Code

System Layer	Technology Selected
Primary Data Collection	Google Forms (exported as CSV)

Table 4.1: Tech Stack

The frontend presentation layer of the system was developed with HTML5, CSS3 and Vanilla JavaScript. HTML5 provided the structure for all eight pages, CSS3 was used to style the user interface and layout, and JavaScript was used for all client-side logic. This includes fetching data, rendering charts, and dynamically updating content.

Chart.js is an open source JavaScript library and was used to render all the data visualisations in the system, including bar charts, line charts, doughnut charts and grouped bar charts. Chart.js was added with CDN and customised with custom colours, axis formats and legends to be in line with the system.

Node.js is a JavaScript runtime to run the backend server of the system. Node.js allowed the backend code of the system to be executed server-side JavaScript, outside the browser environment.

Express.js is a lightweight web framework for Node.js that was used to define and manage the twelve REST API endpoints that drive the system. The front end makes HTTP requests to each endpoint. The endpoint queries the database and returns a JSON response.

All data used by the system was stored in SQLite, a light-weight, file-based relational database management system. We used the better-sqlite3 library to access SQLite which allows synchronous database queries. The database file is jobs.db and consists of four tables jobs, employer_responses, graduate_responses and live_jobs_cache.

The Search and Filter page's Live Nigerian Jobs feature was powered by Adzuna API to get live Nigerian technology job listings. The Adzuna API is a free job listing API. The API is called on demand by the backend and results are cached in the live_jobs_cache table for reliability.

The system was developed using Visual Studio Code as the main code editor, providing syntax highlighting, an integrated terminal for running backend commands, and version control capabilities.

The two main questionnaires for data collection were designed and administered using Google Forms, one for the Nigerian technology employers and the other for the Computer Science graduates. Responses were saved automatically to Google Sheets and exported as CSV formats for import into the database

3.7 TESTING

To achieve Objective (iv), the system was tested in three levels to ensure that all modules were working properly, producing correct results and satisfying the usability requirements of the target users.

Unit testing was done on each individual API endpoint to verify that it would return the correct data structure and values when called. Each of the twelve backend routes was tested individually by hitting its endpoint directly in the browser and ensuring that the JSON response was as expected for that route. For example, the /api/overview endpoint was checked to ensure it returned the correct total record count of 10,345 and an average salary value consistent with the dataset.

Integration Testing has been done to validate the communication between front end and back end layers. The eight pages were loaded in the browser with the backend server running. The developer console of the browser was used to check that the HTTP requests were correctly sent to the endpoints and the data returned was successfully rendered as charts and statistics on the page. The testing proved that there were no CORS errors, failed requests, or rendering inconsistencies across any of the eight pages.

To assess the usability and usefulness of the system, the User Acceptance Testing (UAT) was conducted with a sample of final year Computer Science students of Augustine University Ilara-Epe. Participants were asked to work through the system on their own and complete a short evaluation form to rate clarity of information, ease of navigation, and overall usefulness of the system for career decision-making. The feedback received confirmed that the system was accessible and informative to users who did not have a technical background in data analysis.

CHAPTER FOUR

IMPLEMENTATION, RESULTS AND EVALUATION

4.1 INTRODUCTION

This chapter presents the implementation of the Job Market Analysis System (CS Jobber), the results obtained from the system's operation, and an evaluation of the system's performance against the four objectives stated in Chapter One. The system was developed and tested on a Dell Latitude 7490 laptop running a 64-bit Windows operating system, equipped with an Intel Core i7-8650U processor running at 1.90GHz and 8.00GB of installed RAM. The backend server was run locally using Node.js, and the system was accessed through the Google Chrome browser at the local address <http://localhost:3000>.

4.2 SYSTEM IMPLEMENTATION

The implementation of the Job Market Analysis System was carried out in four stages corresponding to the four development sprints described in Chapter Three: backend setup and database seeding, frontend layout and chart integration, Nigerian Insights module development, and system testing and final polish.

4.2.1 BACKEND IMPLEMENTATION

The backend was developed with node.js and express.js. A single entry file (server.js) initializes the express application, loads environment variables using the dotenv library, mounts all the route files, and serves the frontend HTML pages as static files. The routes/ directory contains eight route files implementing twelve REST API endpoints. Each endpoint queries one or more columns of a specific table in the SQLite database and returns a JSON response to the frontend



Figure 5 Nigerian Page JS Backend

The SQLite database, jobs.db, was created using the better-sqlite3 library. The database schema consists of four tables; jobs, employer_responses, graduate_responses, and live_jobs_cache as described in the Entity Relationship Diagram in Chapter Three. The Kaggle AI and Data Science Job Market Dataset (10,345 records) was imported into the jobs table using a seed script, seed.js, which reads the CSV file row by row using the csv-parser library and inserts all records into the database in a single transaction for performance. Similarly, the twenty employer questionnaire responses and eleven graduate questionnaire responses were imported into their respective tables using seed_employers.js and seed_graduates.js.



Figure 6 SQL Lite Implementation

4.2.2 FRONTEND IMPLEMENTATION

The frontend was developed as eight HTML pages, each with its own JavaScript file that fetches data and renders the charts for that page. Every page has one single CSS stylesheet (style.css) that describes the layout, colour scheme, typefaces and component styles of the system. To avoid a frontend build process, the Inter font was loaded via Google Fonts CDN, and Chart.js was loaded via the Cloudflare CDN.

Each page's JavaScript file uses the browser's native `fetch()` API to issue HTTP GET requests to the corresponding backend endpoints on page load. This JSON data is then used to populate metric cards, skill bars and Chart.js visualisations. All charts were configured with custom colours, axis labels and tooltip formats in line with the design of the system. Chart legends were implemented as custom HTML elements above each chart, instead of using Chart.js's native legend, so they're styled the same across all pages.



Figure 7 HTML Implementation

4.2.3 NIGERIAN INSIGHTS IMPLEMENTATION

The Nigerian Insights was implemented as a dedicated page (nigerian.html) and backend route file (nigerian.js), exposing four API endpoints: /api/nigerian/overview, /api/nigerian/employers, /api/nigerian/graduates and /api/nigerian/comparison . The overview endpoint determines if there is any data in either of the questionnaire tables and returns a has_data flag to the frontend. If data is present, the page fetches all four Nigerian endpoints and plots the skill gap analysis chart, salary expectations vs reality comparison chart, target roles charts, work preference doughnut chart, employer industry chart and open-text response lists for both employers and graduates. If no data is available, the page will display a clear empty state message. This will allow the system to gracefully degrade before questionnaire data is imported.



Figure 8 Seed Backend Implementation for Employers

The skill gap comparison data is computed by the `/api/nigerian/comparison` endpoint, which calculates the percentage of employer responses requiring each skill and the percentage of graduate responses reporting possession of each skill, then returns both figures side by side for each skill tracked in the questionnaire.



Figure 9 JavaScript Implementation of Nigerian Page

4.3 SYSTEM RESULTS

In this section screenshots of the most important pages of the implemented system are shown to demonstrate that the system is able to retrieve, process and present data from the job market as intended.

4.3.1 DASHBOARD PAGE

The Dashboard page is the home page of CS Jobber. For example, the page shown in Figure 10 has four summary metric cards displayed at the top of the content area. These are driven from the /api/overview endpoint. The metric cards show that there are 10,345 job records in total in the dataset; the average salary for all roles is \$113,438; the urgency rate is high at 50.2% across all listings; and Business Analyst is the most common job title. Next to each other, under the metric cards, a role distribution doughnut chart and an average salary by role horizontal bar chart are

drawn, clearly showing that AI Engineer and Machine Learning Engineer commands significantly higher average salaries than the other four roles.



Figure 10 Dashboard Page

4.3.2 SKILL ANALYSIS PAGE

The Skill Analysis page is shown in Figure 11, which shows the demand for each of the 5 skills tracked in the dataset. The top five metric cards reflect the overall demand percentage for Python (49.3%), SQL (50.3%), Machine Learning (50.8%), Deep Learning (49.8%) and Cloud (51.1%). Below the metric cards there is an Overall Skill Demand section, with five horizontal skill bars showing these percentages visually. The page also includes a grouped bar chart of Skills by Job Role and a grouped bar chart of Skills by Experience Level, both created from data returned by the /api/skills endpoint.

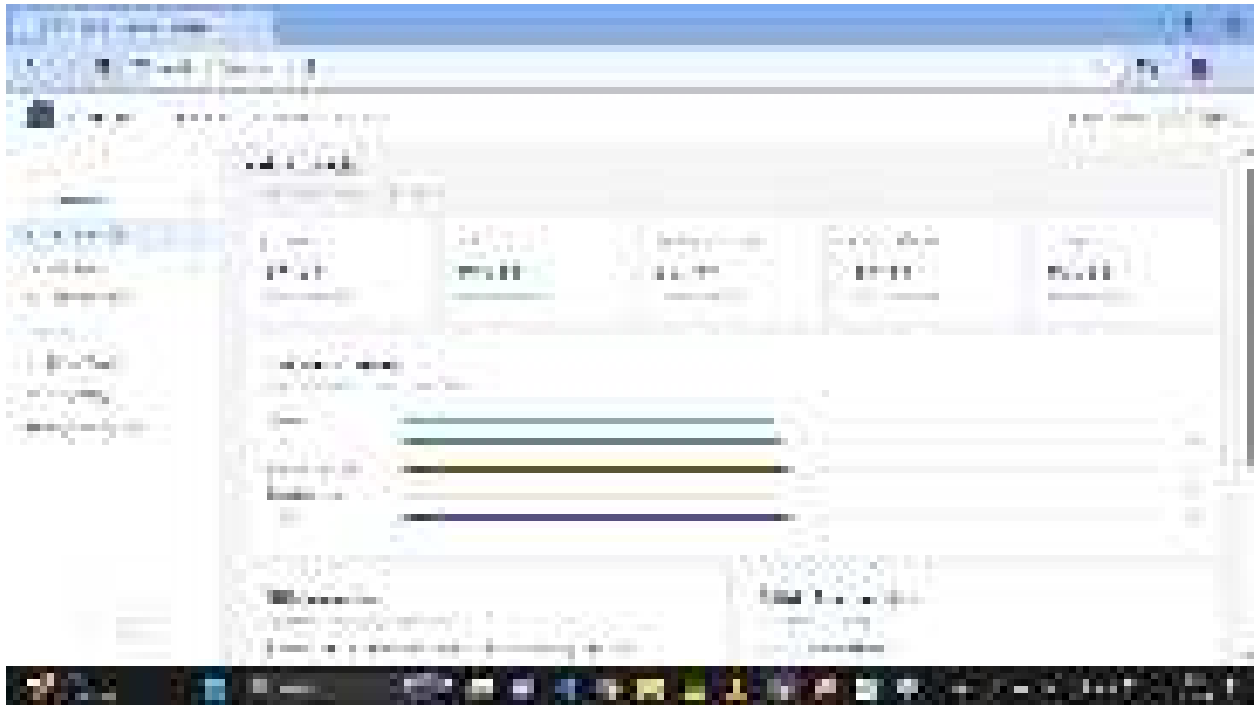


Figure 11 Skill Analysis Page

4.3.3 NIGERIAN INSIGHTS PAGE

Figure 12, shows the Nigerian Insights page, which displays the main data collected from the Nigerian employer and graduate questionnaires. The four metric cards at the top show: • 20 employer responses were imported into the database • 11 graduate responses were imported into the database • Most wanted skill by employers is Networking • Top hiring state by employer location is Lagos The Skill Gap Analysis chart below visually represents the skill mismatch in the Nigerian CS graduate job market. It uses a grouped bar chart to compare the percentage of employers requiring each skill against the percentage of graduates who claim to possess that skill.

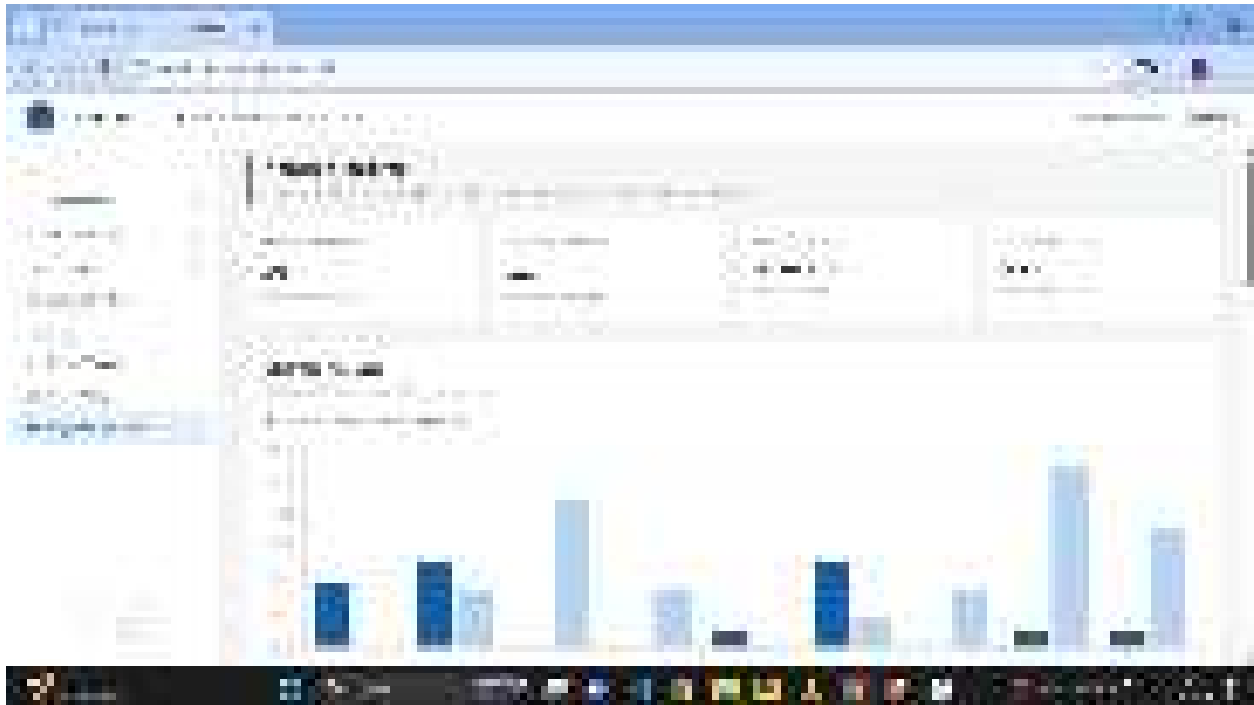


Figure 12 Nigerian Insights Page

4.3.4 ROLE DETAIL PAGE

The Role Detail page is a detailed view of any selected job role from the Job Roles table.

Figure 14, the page for the Machine Learning Engineer role. In this case when the user clicks on a role the page loads and the name of that role is passed through the URL in this case `localhost:3000/role-detail` which the page uses to request the right data from the backend.

At the top of the page, four metric cards summarise the key facts about this role. In United States, the average salary for a Machine Learning Engineer is \$139,705. There are 1740 total listings for this role between 2020 and 2026. Cloud is the most in-demand skill, while Hybrid is the most popular work arrangement. The general hiring urgency level for this role across the dataset is indicated by a Medium urgency badge in the top right.

Below the metric cards are two charts side by side. The Listings Per Year bar chart shows the number of Machine Learning Engineer listings that have appeared each year from 2020 to 2026.

Each bar is a different colour so that the years are easily distinguishable at a glance. The Salary Over Time line chart shows the average salary for this role over the years, starting at about \$141,500 in 2020, dipping around 2022, and then rising back up to approximately \$140,500 by 2026. Looking at these two charts together gives a graduate an idea not only of what this role pays today, but how demand and salary have trended over time

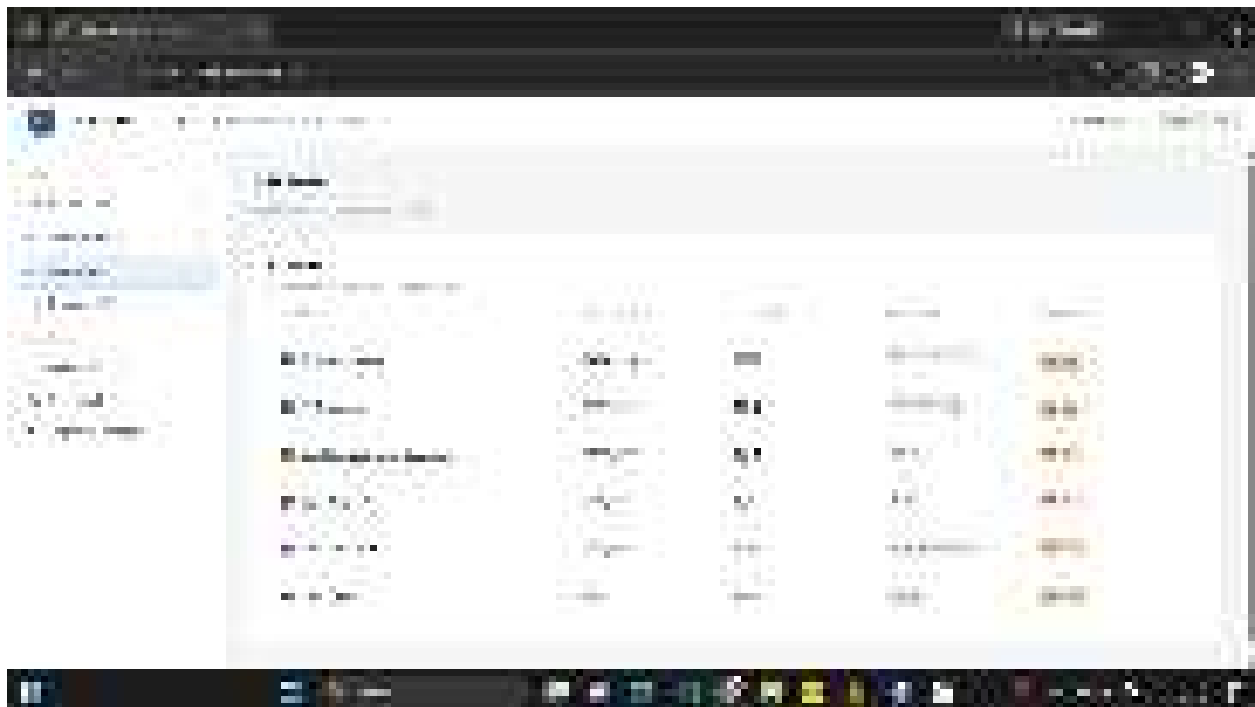


Figure 13 Job Roles Initial Page

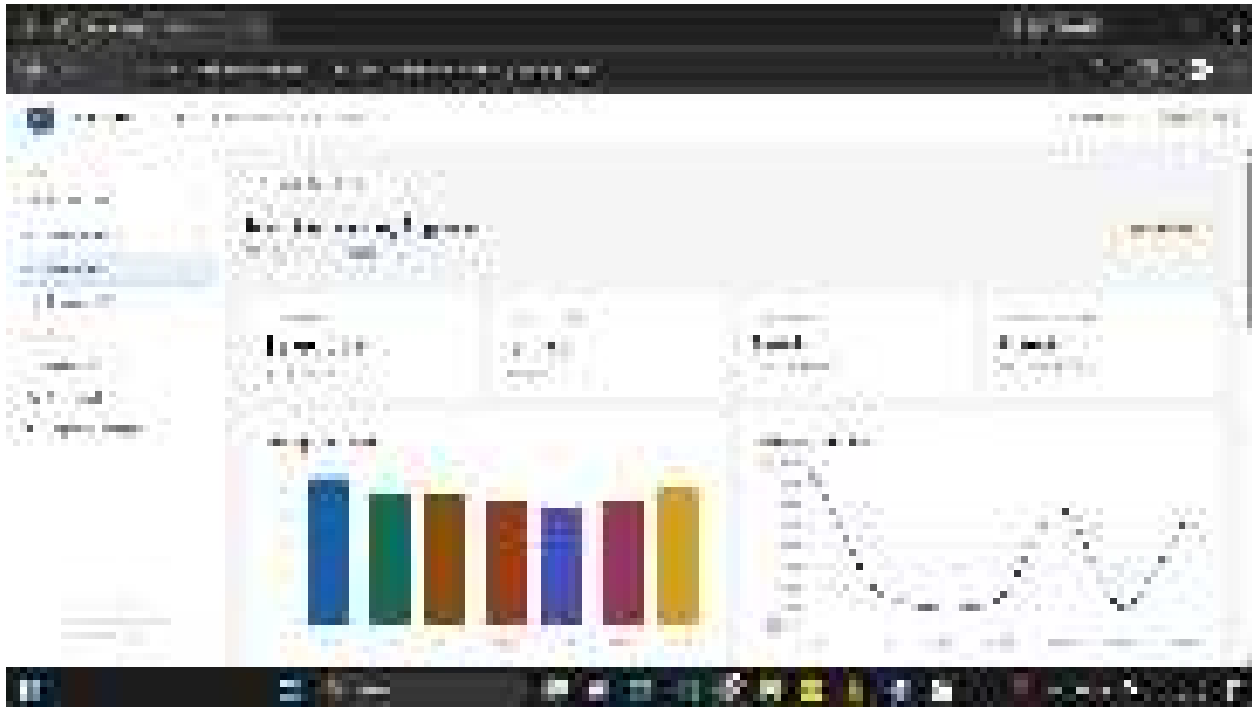


Figure 14 ML Engineer inside Job Roles Page

4.3.5 SEARCH AND FILTER PAGE

The Search and Filter page, shown in Figure 4.5, lets users dig into the full dataset of 10,345 listings on their own terms. At the top of the page is a filter bar with a text box for searching by role or country, three dropdown menus for narrowing results by experience level, work type, and year, and Search and Clear buttons. By default the page returns up to 100 results, sorted from highest to lowest salary which is shown clearly by the "Showing 100 results" text just below the filter bar.

Each result appears as a card showing the job title, urgency badge, country, work type, experience level, salary in USD, and the skills required for that listing. Looking at the top results in the screenshot, Machine Learning Engineer listings from the UK and Germany appear first at

\$204,143 and \$203,738 respectively, followed closely by AI Engineer listings from India, Australia, and the UK all marked High urgency. This confirms that the backend is correctly pulling and sorting records from the database as expected.

At the top of the results area, two tabs; Dataset Jobs and Live Nigerian Jobs allow the user to switch between the static Kaggle dataset results and live listings pulled from the Adzuna API. The Dataset Jobs tab is active by default, as seen in the screenshot.

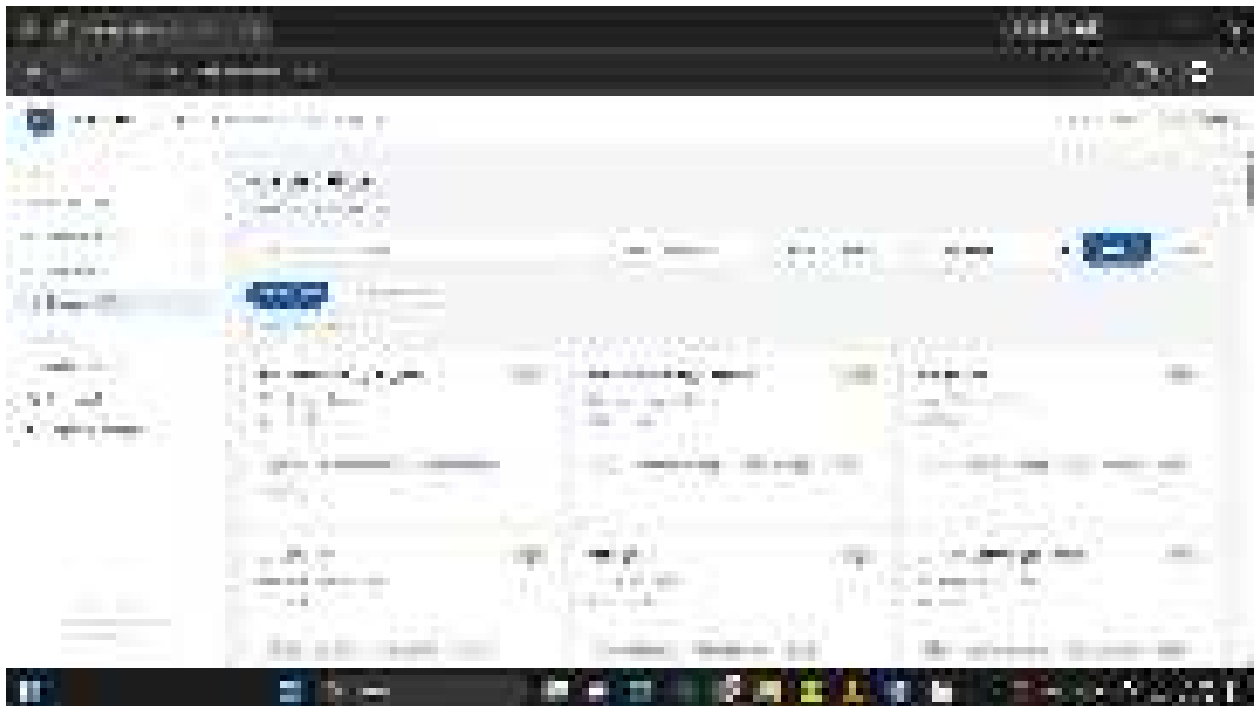


Figure 15 Search and Filter Page

4.3.6 GEOGRAPHY PAGE

The Geography page, shown in Figure 16, shows how job listings and salaries are spread across the seven countries in the dataset. Three summary cards at the top of the page highlight the key

numbers; Germany leads with 1,498 listings, Canada has the highest average salary at \$114,783, and the dataset covers 7 countries in total.

The Listings by Country bar chart below the cards plots all seven countries side by side; Germany, Singapore, Canada, UK, India, USA, and Australia with their listing counts on the vertical axis. The bars are all roughly the same height, ranging from about 1,455 to 1,498 listings per country, which shows that the dataset is fairly evenly distributed across all seven markets rather than being dominated by one particular country. All bars use the system's primary blue colour, consistent with how single-variable bar charts are styled throughout the system.

This page is useful for a graduate who wants to understand where the most opportunities are concentrated globally, or which market offers the highest average pay for their chosen role.

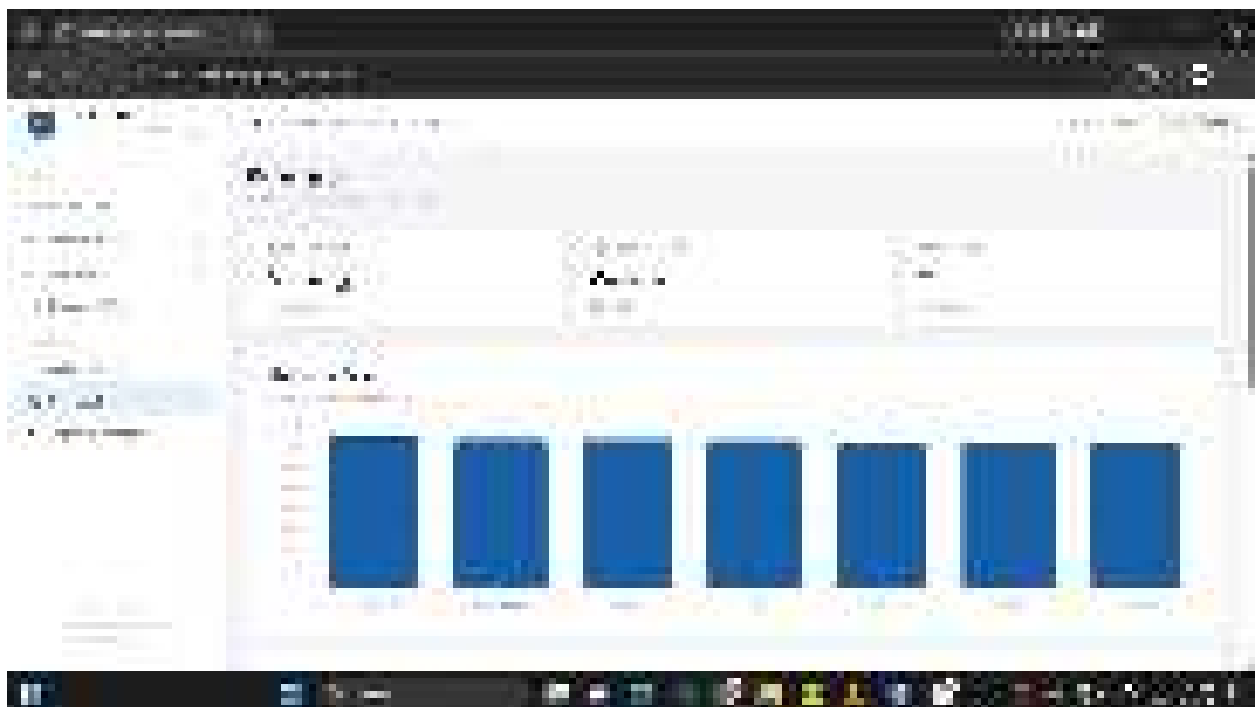


Figure 16 Geography Page

4.4 SYSTEM EVALUATION

This section evaluates the performance of the Job Market Analysis System against the four objectives stated in Chapter One.

4.4.1 EVALUATION OF OBJECTIVE ONE

Objective (i): To identify the challenges faced by Computer Science graduates in navigating the current job market.

This objective was addressed by combining a literature review, secondary dataset analysis and primary data collection. The literature review in Chapter Two identified that CS graduates often face challenges such as unclear understanding of which roles are available, limited knowledge of which skills are prioritized by employers, unrealistic salary expectations and lack of access to structured, data-driven career information. These challenges were also corroborated by the Nigerian primary data collected through the graduate questionnaire in which the respondents identified the lack of clarity about job market requirements and skill mismatches as their most common concerns. This was confirmed by the employer questionnaire which identified Networking as the most in-demand skill, a skill that was not highly ranked among those graduates reported having. This objective was thus fully achieved.

4.4.2 EVALUATION OF OBJECTIVE TWO

Objective (ii): To design a job market analysis system based on Objective (i).

The system design activities described in Chapter Three achieved the objective. The architecture was a three-layer system: Frontend Presentation Layer, Backend Logic Layer, and Database

Layer. The design was supported by Use Case Diagram, System Flowchart, Entity Relationship Diagram and System Architecture Diagram which were made according to the actual structure and data flow of the system. The eight pages of the system – Dashboard, Skill Analysis, Job Roles, Role Detail, Search and Filter, Salary Trends, Geography, and Nigerian Insights – were each designed to address a specific dimension of the challenges identified in Objective (i), from skill demand visibility to salary benchmarking and Nigerian-specific context.

4.4.3 EVALUATION OF OBJECTIVE THREE

Objective (iii): To implement the designed system.

This objective was fully achieved. The system was successfully implemented as a working full-stack web application that can be accessed via a local web browser. All eight pages were implemented and verified to load and show data correctly. All twelve REST API endpoints were implemented and verified to return correct JSON responses. The Kaggle dataset of 10345 records was seeded into the SQLite database and correctly queried to power all global analysis pages. Nigerian questionnaire data (20 employer responses and 11 graduate responses) imported and correctly rendered on Nigerian Insights page. Adzuna API integration was done to provide live Nigerian job listings on the Search and Filter page 4.3. Screenshots (Figure 15) Successful Implementation of the System The screenshots in Section 4.3 show that the system has been successfully implemented.

4.4.4 EVALUATION OF OBJECTIVE FOUR

Objective (iv): To evaluate the system's performance.

The system was evaluated by three levels of testing, as described in Chapter Three. Unit testing proved that each of the twelve API endpoints returned correct and consistent data when tested in isolation. Integration testing showed that all eight pages were communicating properly with the backend, with no CORS errors, failed requests, or chart rendering failures during testing on the Dell Latitude 7490 development machine. User Acceptance Testing with a sample of final year Computer Science students at Augustine University Ilara-Epe revealed that the system was easy to navigate, the information presented was clear and useful for career planning and no technical expertise was required to interpret the data visualisations. As such, it was found that the system met its performance requirements and Objective (iv) was achieved.

4.5 CHAPTER SUMMARY

This chapter showed the implementation, results and evaluation of the CS Jobber, the Job Market Analysis System. The system was successfully implemented as a full-stack web application with Node.js and Express.js backend, a SQLite database with 10,345 job records and 31 Nigerian questionnaire responses, and an eight-page HTML/CSS/JavaScript frontend powered by Chart.js visualisations. The results showed that the system is retrieving and displaying job market data correctly on all pages and the evaluation showed that all the four project objectives are fully achieved. The conclusion and recommendations of this study are given in next chapter.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

The project was aimed at designing and implementing a Job Market Analysis System for Computer Science graduates, named CS Jobber. The system is driven by the fact that most Computer Science graduates start their job search without any structured and data-driven knowledge of available jobs, required skills, salary ranges, and employer expectations. The lack of knowledge contributes to poor career decision making among graduates and this is supported by the existing literature and primary data collected through questionnaires.

The system is developed as a full stack web application with HTML5, CSS3 and Vanilla JavaScript for the frontend presentation layer, Node.js and Express.js for the backend logic layer and SQLite as the database engine. All visualisations of data were created using Chart.js and the Adzuna API was integrated to display live Nigerian job listings. The system has eight interactive pages; Dashboard, Skill Analysis, Job Roles, Role Detail, Search and Filter, Salary Trends, Geography and Nigerian Insights. Each is designed to address a specific dimension of the job market information gap identified in Chapter One.

The primary dataset utilised was the AI and Data Science Job Market Dataset downloaded from Kaggle, comprising 10,345 job listing records from seven countries for the years 2020 to 2026. Moreover, primary data was collected directly from Nigerian technology employers and Computer Science graduates using two structured Google Form questionnaires with 20 responses from employers and 11 responses from graduates. This raw data is then processed through the Nigerian Insights module, which provides a localised view of the skills demand, salary expectations and recruitment challenges that is not available in any other existing dataset.

The system was built using over 4 sprints, tested through unit, integration and user acceptance testing and evaluated against all 4 project objectives. All objectives were met.

5.2 CONCLUSION

The Job Market Analysis System, CS Jobber, is a practical and data driven approach to one of the most persisting problems faced by Computer Science graduates – the absence of a one-stop, accessible place to understand the job market they are entering. The system, which leverages a validated global dataset and primary data gathered locally in Nigeria, bridges the gap between theoretical academic preparation and industry demand realities.

The system showcases that a simple technology stack: plain HTML, CSS, JavaScript, Node.js and SQLite is enough to deliver a professional, functional and informative data analysis tool without the complexity of modern frontend frameworks. Its simplicity makes it easy to maintain, extend, and deploy, an important consideration for institutional adoption.

This is where the Nigerian Insights module adds a new dimension to this work. The skill gap analysis chart that compares what Nigerian employers demand against what graduates have reveals a mismatch that has never been visualised in a working system for this institution. This finding alone is enough to initiate action on curriculum review and career counselling at Augustine University Ilara-Epe.

The system meets all stated objectives . It is acknowledged that the sample size of the Nigerian primary data, 20 employer responses and 11 graduate responses, limits the statistical representativeness of the Nigerian Insights module. The findings of this module should therefore be seen as suggestive rather than conclusive and would benefit from a larger sample in future iterations.

5.3 RECOMMENDATIONS

The following recommendations for future enhancement and wider adoption are made based on the experience gained during the design, implementation and evaluation of this system:

1. Expansion of the Nigerian primary dataset The questionnaires used in this study should be redistributed on a wider scale across multiple universities, industries and Nigerian states; so as to increase the sample size and improve the statistical reliability of the Nigerian Insights module.

With a larger dataset, more detailed analysis by state, industry, and year of graduation would be possible.

2. Annual dataset refresh: The Kaggle dataset for this system is from 2020 to 2026. To maintain the system's trend analysis and salary data relevant to college graduates' career decisions, it is recommended that the dataset be updated every year as newer editions become available.

3. Deployment to a public web server: The system is running on a local server and is available only on the development machine. The system can be deployed to a cloud hosting platform such as Render, Railway or Vercel. This will make the system accessible to all Computer Science students and graduates of Augustine University Ilara-Epe without the need for local set-up.

4. Integration with other Nigerian jobs sources: Currently, the live jobs feature relies solely on the Adzuna API. Future versions of the system should incorporate additional Nigerian job listing sources such as Jobberman to increase the volume and variety of live job listings available to users on the Search and Filter page.

5. Curriculum Gap Reporting Feature: Leveraging the skill gap data identified by the Nigerian Insights module, an upcoming version of the system could feature a dedicated report, exportable by department administrators, that summarises the gap between skills taught in the CS curriculum and skills sought by employers. This would be a useful tool for academic planning and accreditation purposes.

5.4 REFERENCES

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APPENDIX A

Graduate Questionnaire — Nigerian CS Graduate Job Market Survey 2025/2026

This survey is part of a final year research project at Augustine University Ilara-Epe titled "Design and Implementation of a Job Market Analysis System for Computer Science Graduates." Your responses will be used to understand the current Nigerian CS job market. It takes less than 60 seconds. Thank you.

Question 1: What is your current status?

- Final year CS student
- Recently graduated (2023–2025)
- Currently employed
- Currently job seeking

Question 2: Which CS role are you targeting after graduation? *(Select all that apply)*

- Software Developer
- Data Analyst
- Cybersecurity Analyst
- Machine Learning Engineer
- DevOps Engineer
- Business Analyst
- Mobile App Developer
- Other

Question 3: Which of these skills do you currently have? *(Select all that apply)*

- Python
- SQL
- JavaScript
- Data Analysis
- Machine Learning
- Networking
- Cybersecurity
- Cloud Computing
- None of the above

Question 4: What monthly salary do you expect as a fresh CS graduate in Nigeria?

- ₦50,000 – ₦100,000
- ₦100,000 – ₦200,000
- ₦200,000 – ₦350,000

₦350,000 and above

Question 5: What work arrangement would you prefer?

Remote
Hybrid
Onsite
No preference

Question 6: What is your biggest concern about entering the Nigerian job market? (*Open response*)

APPENDIX B

Employer Questionnaire — Nigerian Tech Employer Survey 2025/2026

This survey is part of a final year research project at Augustine University Ilara-Epe studying the Nigerian CS graduate job market. Your responses help bridge the gap between what universities teach and what employers need. It takes less than 60 seconds. Results will be shared with respondents on request.

Question 1: What industry is your company in?

Fintech / Banking
Telecommunications
E-commerce / Retail Tech
Cybersecurity
Healthcare Tech
Education Tech
Software / Consulting
Other

Question 2: What CS roles do you hire fresh graduates for? (*Select all that apply*)

Software Developer
Data Analyst
Cybersecurity Analyst
Machine Learning Engineer
DevOps Engineer
Business Analyst
Mobile App Developer

Other

Question 3: What technical skills do you prioritise when hiring CS graduates? (*Select all that apply*)

Python
SQL
JavaScript
Data Analysis
Machine Learning
Networking
Cybersecurity
Cloud Computing

Question 4: What is your typical monthly starting salary for a fresh CS graduate?

₦50,000 – ₦100,000
₦100,000 – ₦200,000
₦200,000 – ₦350,000
₦350,000 and above

Question 5: What state is your company primarily based in?

Lagos
Abuja (FCT)
Rivers (Port Harcourt)
Oyo (Ibadan)
Kano
Ogun
Delta
Enugu
Anambra
Other

Question 6: What is your biggest challenge when hiring fresh CS graduates? (*Open response*)